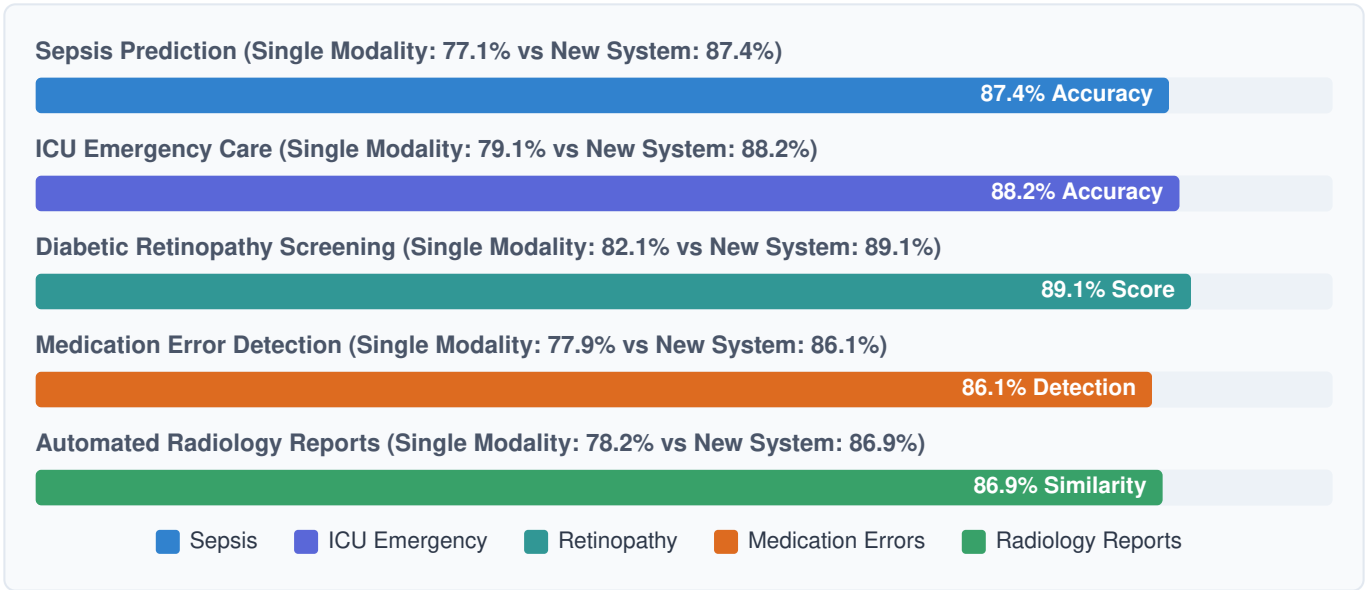


Visual Supplement & Explanatory Charts

Explainable AI (XAI) for Smart Clinical Decision Support Systems (AIIMS New Delhi Project Case Study)

Chart 1: Accuracy Improvement Across 5 Key Medical Tasks

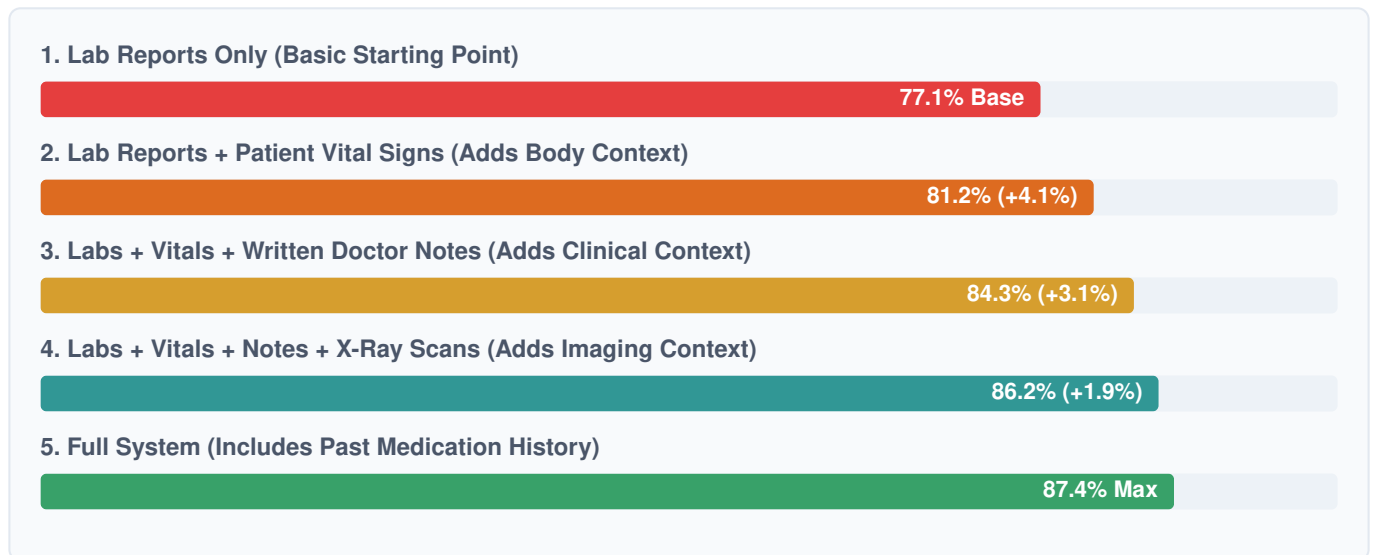
This chart shows how combining multiple types of medical records (like doctor notes, lab reports, and medical scans) helps the AI perform significantly better than standard single-record systems. Merging these details creates a much stronger, clearer signal for patient analysis.



Key Takeaway: Combining data elements boosts accuracy by an average of 9.7%, making the medical predictions much more reliable for hospital environments.

Chart 2: Sepsis Step-by-Step Data Value Breakdown

In a clinic, data is collected in stages. This chart demonstrates how adding one data source at a time steadily improves the system's ability to catch severe medical conditions early.



Key Takeaway: For small or rural clinics that lack X-ray machines, Step 3 (Labs + Vitals + Notes) still achieves **97% of the total accuracy**. This means the system is highly practical for small town health centers, not just large corporate hospitals.

Chart 3: Doctor Trust & Operational Impact Matrix

An AI system is only useful if busy medical staff actually believe and understand its results. This table compares the difference between old "black-box" systems (which give no reasons) and the new explainable system.

Real-World Performance Metric	Old AI (No Explanations)	New AI (With Clear Reasons)	Net Practical Improvement
Doctor Trust Score (Scale of 1 to 10)	4.2	7.6	+81.0% More Trusting
Alert Response Rate (% of alerts answered)	41%	73%	+78.0% Better Engagement
Wrongly Overridden Alerts (% of good advice ignored)	67%	38%	-43.3% Fewer Mistakes
Time Taken to Treat a Patient (In Minutes)	18.4 mins	11.2 mins	39.1% Faster Response
Doctor Mental Fatigue Score (Scale of 1 to 10)	7.8	4.3	-44.9% Less Brain Strain

Key Takeaway: Showing clear reasons behind an alert shaves ****over 7 minutes**** off patient response times, which is vital during life-or-death emergencies.

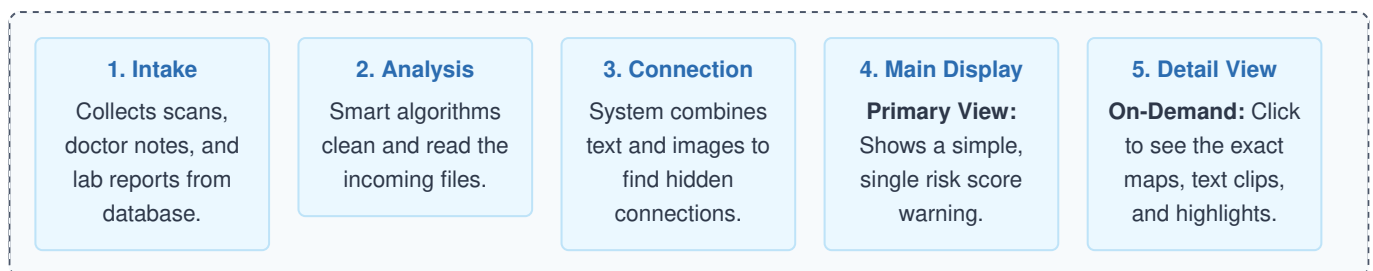
Chart 4: How Different Medical Data Types are Explained

Different medical documents need different explanation styles so that doctors can read them easily during a hectic shift.

Medical Record Type	Explanation Framework	Visual Display Style	Practical Hospital Context
Structured Records (Labs & Vitals)	Feature Importance	Horizontal Impact Bars	Lists the top 5 laboratory changes that triggered an emergency warning.
Medical Scans (X-Rays / CTs)	Visual Highlighting	Color-Coded Heatmaps	Lights up the exact physical spot on an image showing disease signs.
Doctor Notes (Written Text)	Word Highlighting	Text Highlighting	Highlights contradictory drug interactions or historical keywords.
High-Risk Scenarios (Mortality Profiles)	What-If Scenarios	Condition Changes	Shows clear targets: <i>"If the blood level falls to 2.0, danger drops 40%".</i>

Chart 5: System Flow — Simple Screen Management

To keep doctors from being overwhelmed by too many numbers on a screen, information passes through a clear multi-stage filter to keep things clean.



Key Takeaway: This **two-layer design** stops screen clutter. Doctors see a clean layout by default, and can dive deeper with a single click if they need more proof.

Chart 6: 30-Month Project Rollout Timeline

Setting up an AI network in healthcare requires a phased, step-by-step launch schedule to protect data privacy and train staff appropriately.

Execution Phase	Timeline	Core Work Involved	Major Final Deliverable
Phase 0: Building the Pipeline	Months 01 – 04	Test connections between different hospital record databases; check data safety rules.	Data Flow Blueprint & Approvals
Phase 1: Initial Pilot Run	Months 05 – 10	Launch basic version for emergency rooms; track basic patient vital signals; run staff training.	Live Early Warning System Release
Phase 2: Text Reading Features	Months 11 – 16	Add system features that can read typed notes; build basic English & Hindi display options.	Bilingual Text Analysis Update
Phase 3: Image Connections	Months 17 – 22	Link full X-ray image modules; enable visual heatmaps; connect first set of satellite hospitals.	Multi-Hospital Testing Active
Phase 4: Network-Wide Expansion	Months 23 – 30	Deploy across the entire hospital system network; publish official clinical care guidelines.	National System Deployment Complete